Phase Manifold Model: First Principles and Mathematics

Event Geometry, Persistent Record, and Relational Clock Gravity

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Abstract

PMM proposes an underlying event architecture connecting quantum mechanics, general relativity and information theory through Before, During and After. Contact actualizes an admissible interaction; anti-contact marks its completion. Phase readouts organize potential content, realized geometry and retained information. Space is emergent relational motion, depth relates two internal During vantages, and time is angular with action as a common ruler. The phase creases organize the c-order ladder, while light as stillness denotes zero null proper time and the invariant realization reference c. Separate endpoint Afters enter accumulated histories and condition subsequent possibilities and downstream gravity. Investigating force manipulation through depth is a research objective. Definitions, adopted postulates and candidate physical assignments remain distinguished; the technical Core and companions contain the detailed mathematics, recovery conditions and prediction status.

Keywords. Phase Manifold Model; Field Transition Unit; angular time; emergent geometry; persistent record; relational clock; gravity.

Technical companion
The full v3.7.4 manuscript contains the complete core, status classifications, derivations, consistency checks, and research boundary. Read the technical manuscript.

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1. Purpose and Goals

PMM proposes an underlying event architecture connecting quantum mechanics (QM), general relativity (GR), and information theory (IT). It organizes potential, active realization, and completed record as phases of one event. These are PMM correspondences, not claims that the three established theories are identical.

The primary goal is this connecting architecture. The secondary goal is to investigate whether force can be manipulated through depth; this is a research objective, not an established capability. Resolving known physics problems is a further possible outcome of the first two goals.

2. One Complete Event

A Field Transition Unit (FTU) is one complete Before → During → After event:

$$F_{tu} = B \circ D \circ A$$

B = Before = Potential → Quantum Mechanics → Wave → Uncertainty → Pre-spatial

D = During = Realization → General Relativity → Particle → Partial Certainty → Spatial xyz

A = After = Transcription → Information Theory → Cube → Full Certainty → Post-spatial

Uncertainty, partial certainty, and full certainty describe the event-state heuristic: a completed outcome is definite as a record. They do not imply complete knowledge of every microscopic detail. The A cube is a record representation, not active three-dimensional A-space.

3. Contact and Completion

@ = Wave Function Collapse / Contact

−@ = Anti Contact

B → @ → D → −@ → A

In PMM, @ actualizes an admissible pair interaction and opens D. At −@, D reaches terminal closure; C_3 governs transcription into persistent A records. One shared pair interaction is one FTU, with two separate endpoint Afters associated with its participant lineages. They share one event identity and one event-level source energy.

C_1 governs entry, C_2 conditions the realized path, and C_3 determines what persists:

$$\Theta_con = \{C_1[@,\Phi],\ C_2[\Phi,\delta,\Sigma],\ C_3[\Phi,\Sigma]\}$$

Here Φ is accumulated record history, δ is inter-participant distance, and Σ is closure spread. Internal depth z^D is distinct from δ.

4. Readouts Across Before During and After

The readout tuples are the organizing reference for PMM. Each row describes the same event through its B, D, and A phase roles. The entries need not be equal or independent, and the notation alone does not derive a physical evolution law.

Readout	Before B	During D	After A
c-order F_{tu}^{c}	$(c^0)_{mp}$	$(c^1,c^2)c$	$(c^3)k_B$
Spatial F_{tu}^{xy}	$\psi(x,y,z)$	xyz	xy
Time F_{tu}^{T}	$[\hbar/(m_pc^2)]\Delta\theta^B$	d/c	$\Delta\theta^A/\omega^A$
Mass F_{tu}^{M}	m_p	E^D/c^2	$\hbar\omega^A/c^2$
Energy F_{tu}^{E}	m_pc^2	E^D	$\hbar\omega^A$
Phase action	$S^B = \hbar\Delta\theta^B$	$S^D = \hbar\theta^D$	$S^A = \hbar\Delta\theta^A$

The spatial tuple makes the phase distinction explicit:

$$F_{tu}^{xy} = (\psi(x,y,z), xyz, xy)$$

B carries potential spatial content, D realizes spatial geometry, and A retains the completed x-y readout. A retained readout is information about realization, not continuing active space.

The time tuple is $F_{tu}^{T} = (T^B,T^D,T^A)$. Its B and A spans use the candidate assignments $\Delta\theta^B = \arccos\sqrt{P_e@}$ and $\Delta\theta^A = -\ln P_e@$, with $0 < P_e@ \leq 1$ for the selected event. $P_e@$ is the C_1-defined pair-contact probability. The phase-action interpretation of these spans remains candidate; the temporal metric is open.

5. Space Is Emergent Through the Motion Operations Principle

M_{op} — Motion Operations Principle: Space is realized relation, and motion is the evolution of that relation. B carries potential relation and motion; D realizes relation as active spatial geometry and relational motion; A retains the completed outcome as record.

Its kinematic readout organizes separation, separation rate, and change in separation rate:

$$M_{op}(\delta_{ij}) \equiv (\delta_{ij}, d\delta_{ij}/dt, d^2\delta_{ij}/dt^2)$$

Here t is comparison time. This tuple summarizes the readouts; it does not determine their dynamical evolution. An FTU has no isolated physical realization; the multi-FTU perspective describes relational geometry among events and participant lineages.

Within D, the two internal vantages V_1 and V_2 each carry an x-y readout. They are event slots rather than fixed participant identities. Their common scale supplies relational depth:

$$\rho_1 = \sqrt{(x_1^D y_1^D)},\ \rho_2 = \sqrt{(x_2^D y_2^D)}$$

$$z^D = L = \sqrt{(\rho_1\rho_2)}$$

Depth is therefore distinct from ordinary height and width. It is active during D and ends at transcription; there is no active z^A. This distinction motivates the depth-and-force research objective without supplying a force-control law.

6. Time Is Angular and Action Is the Ruler

PMM describes time through phase-specific angular readouts. Action supplies their common measure, in joule-seconds:

$$S^B = \hbar\Delta\theta^B,\ S^D = \hbar\theta^D,\ S^A = \hbar\Delta\theta^A$$

$$F_{tu}^{s} = S^B + S^D + S^A$$

During D, $dS^D = \hbar d\theta^D$. The adopted closure is $\theta^D = \pi$, giving $S^D_cl = \pi\hbar$. Common scale and vantage imbalance evolve as:

$$L = (\hbar c/E^D)\theta^D,\ r = r_0\cos(\theta^D/2)$$

Terminal closure requires r = b = Σ = 0; the shape-mismatch mode b has no adopted full evolution law. If $T^D = d/c$ is identified with action time S^D/E^D, then d = L follows conditionally. Angular progression, proper time, and comparison time must retain their stated distinctions.

7. The Ladder Creases and Light as Stillness

$$c^0 \to c^1 \to c^2 \to c^3$$

$$F_{tu}{}^c = ((c^0)_{mp},(c^1,c^2)c,(c^3)k_B)$$

The constants act as phase creases: m_p anchors entry, c scales realization, and k_B converts the information readout into entropy units. Their accepted physical values are inputs; their architectural assignments are PMM's proposal. Both c^1 and c^2 belong to D. The native event ladder ends at c^3.

Light as stillness: c is the invariant still reference of D. Light's null propagation has zero proper time; this does not supply a photon rest frame or prevent angular D progression. The internal scale/action-time relation $L = cT^D_action$ follows on its stated branch, rather than defining every laboratory separation or interaction duration.

8. Separate Afters and Persistent Information

C_3 transcribes a completed event into two lineage-endpoint records:

$$R^A_i|\alpha \text{ and } R^A_j|\alpha$$

Both refer to event α and its one event-level energy. The current minimal numerical record is $R^A_\alpha = \{\omega^A,a_cl,I^A\}$, with $I^A = -\ln P_e@$. Frequency and terminal planar shape support reconstruction of completed x-y scale and shape. They do not by themselves establish retention of an arbitrary wavefunction or a full image sequence.

The adopted frequency transcription and $E = \hbar\omega$ give $E^A = E^D$ within the same event. Completed records enter their respective lineage histories Φ. Entropy-unit readout $k_B\, I^A$ must be distinguished from a derived thermodynamic entropy-production law.

9. Angular Recursion and Downstream Gravity

The Angular Temporal Recursion Principle is a 3.7.4 extension first-principle postulate:

$$A\alpha(c^3) \to \Phi \to B\beta(c^0), \text{ with } \alpha \text{ causally prior to } \beta$$

Completed histories contribute to subsequent possibilities. Coherent contributions can interfere when a phase-preserving channel is supplied. The native generation map $\mathcal{G}\theta$ and phase-preservation mechanism remain open. Video playback illustrates conditional record-driven preparation using present apparatus and energy; it does not require reversing physical time.

Gravity belongs to the relational multi-FTU sector: accumulated records condition D clock response and future interactions. Its current readout is:

$$\kappa = (d\theta^D/dt)/(d\theta^D/dt)_0,\ \ \varphi = -\ln\kappa$$

$$F_{tu}{}^G = \nabla\varphi,\ \ g = c^2\nabla\varphi$$

The source extraction and dynamical bridges remain separately identified in the Core. Detailed comparisons with established physics belong in Recovery; independently specified prospective tests belong in Predictions.

Technical source: PMM 3.7.4 Core v71.34.

Companions: Recovery v1.19 and Predictions v1.24.